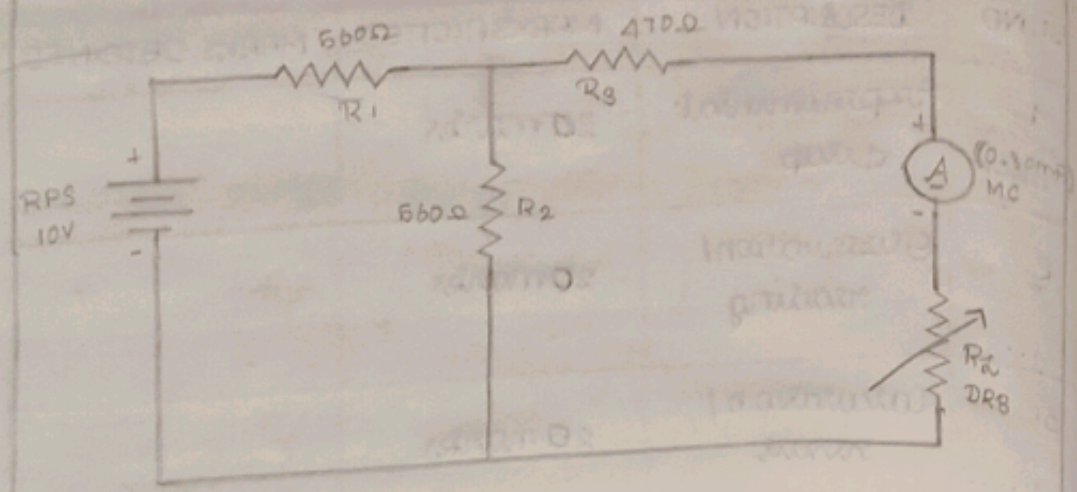
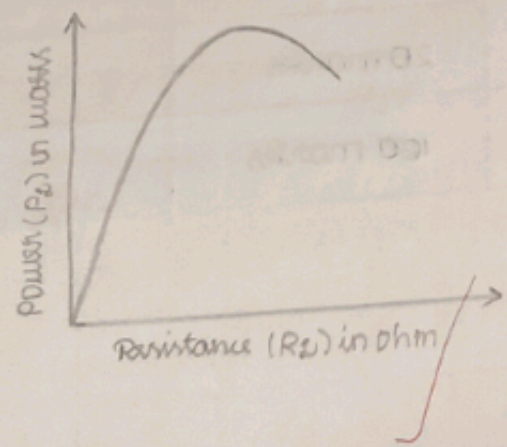
CIRCUIT DIAGRAM : MAXIMUM POWER TRANSFER THEOREM



MODEL GRAPH:



Exp no: 06	VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM
Date: 21/02/2026	

AIM:

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

APPARATUS REQUIRED:

SI.NO	NAME OF THE APPARATUS	RANGE / RATING	QUANTITY
1	Voltmeter	(0-15V) MC	1
2	Ammeter	(0-500mA) MC	1
3	Resistor	560Ω	2
		470Ω	1
4	RPS (DC supply)	15V	1

THEORY:

The maximum power transfer theorem states that "A load will receive maximum power from a linear bilateral DC network when its total resistive value is equal to the Thevenin resistance of the network as seen

TABULAR COLUMN 1 :

For theoretical calculation : $R_{TH} = 750\Omega$

Sl.NO.	LOAD RESISTANCE (R_L) in ohm	LOAD CURRENT (I_L) in mA	LOAD VOLTAGE (V_L) in volts	LOAD POWER (P_L) in watts.
1.	750Ω	3.33 A	2.49V	8.39 mW

TABULAR COLUMN 2 :

For practical calculation:

Sl.NO.	LOAD RESISTANCE (R_L) in ohm	LOAD CURRENT (I_L) in Ampere	LOAD VOLTAGE (V_L) in volts	LOAD POWER (P_L) in watts.
1.	330	4.7×10^{-3}	1.551	7.29×10^{-3}
2.	470	4.20×10^{-3}	1.974	8.29×10^{-3}
3.	800	3.3×10^{-3}	2.64	8.71×10^{-3}
4.	1k	2.9×10^{-3}	2.9	8.41×10^{-3}
5.	2k	1.86×10^{-3}	3.72	6.92×10^{-3}
6.	3k	1.37×10^{-3}	4.11	5.63×10^{-3}

THEORETICAL CALCULATIONS :

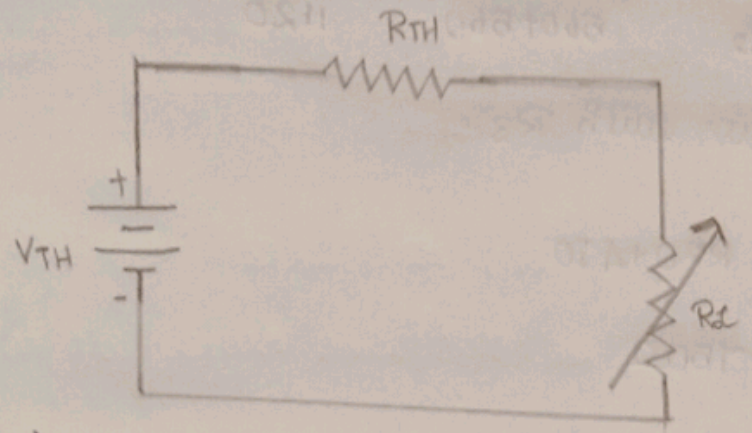
$V_S = 10V$

By VDR $\Rightarrow V_{th} = V_S \times \frac{R_2}{R_2 + R_1}$

$V_{TH} = 10 \times \frac{560}{1120}$

$V_{TH} = 5V$

For the network in figure (a), maximum power will be delivered to the load when $R_L = R_{TH}$.



In a simpler form the circuit may contain a voltage source Vs having internal resistance Rs and connected across a load RL to find power.

The maximum power transfer theorem tell us that the load should be equal in magnitude to the source resistance for maximum power to be absorbed by the load.

PROCEDURE :

1. Make connections as per the circuit diagram.

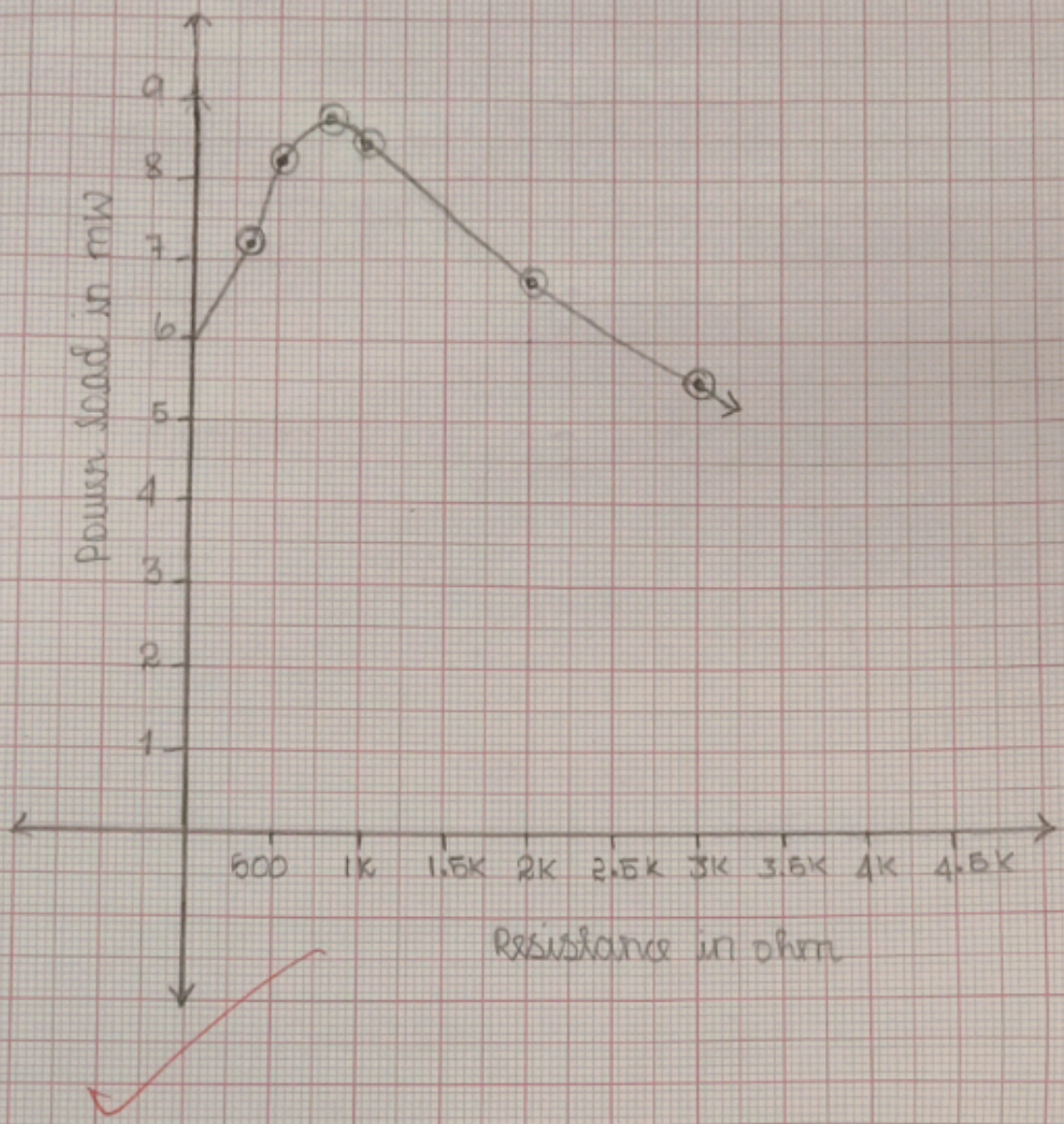
2. Change the resistance RL whose value close to R_{th}, measure the corresponding V_L, I_L and calculate the power P_L and enter into the table (2).

3. Plot a graph between RL and P_L and find the RL corresponding to maximum power transfer.

4. Verify the measured values of RL at maximum power transfer as same as calculated and found graphically.

Scale:

In x axis - 1 unit - 500Ω
In y axis - 1 unit - 1mW



$R_{TH} = R_1$ and R_2 is parallel

$$= \frac{R_1 \times R_2}{R_1 + R_2} = \frac{560 \times 560}{560 + 560} = \frac{313600}{1120} = 280\Omega$$

This is in series with R_3:

$$R_{equ} = 280 + 470$$

$$= 750\Omega$$

Maximum power transfer:

$$P = \frac{(V_{TH})^2 R_L}{(R_{TH} + R_L)}$$

$$R_L = R_{TH}$$

$$P = \frac{V_{TH}^2}{4R_L}$$

$$R_L = 750\Omega$$

$$P = \frac{5^2}{4 \times 750} = 8.33\ mW$$

$$I_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{5}{1500}$$

$$I_L = 0.0033A$$

$$I_L = 3.33mA$$

$$V_L = I_L \times R_L$$

$$V_L = 3.33 \times 10^{-3} \times 750$$

$$V_L = 2.49V$$

MARKS SPLIT-UP:

SI NO	DESCRIPTION	MARKS ALOTTED	MARKS OBTAINED
1.	Experimental setup	20 marks	20
2.	Observation / reading	20 marks	20
3.	Calculation / result	20 marks	20
4.	Viva	20 marks	19
5.	record	20 marks	19
6.	Total	100 marks	98

RESULT:

Thus maximum power transfer theorem is verified practically and theoretically.